

Chemistry: Organic Chemistry

Course Information

Course Name: Chemistry – Organic Chemistry

Level: Beginner / Intermediate

Prerequisites: Basic chemistry and chemical bonding

Estimated Duration: 5–6 Hours

Course Description: Organic chemistry is the study of carbon-containing compounds, their structures, properties, reactions, and applications. This course introduces hydrocarbons, functional groups, nomenclature, isomerism, major organic reactions, and polymers.

Learning Outcomes

Explain why carbon forms many compounds.

Identify major types of hydrocarbons.

Distinguish between alkanes, alkenes, and alkynes.

Identify common functional groups.

Name simple organic compounds.

Understand basic isomerism.

Explain common organic reactions.

Describe polymers and their applications.

Identify applications of organic chemistry in everyday life.

Lesson 1: Introduction to Organic Chemistry

Organic chemistry is the branch of chemistry concerned mainly with carbon-containing compounds.

Carbon can form strong covalent bonds with carbon, hydrogen, oxygen, nitrogen, sulfur, and halogens.

Catenation is the ability of carbon atoms to bond with other carbon atoms. Carbon has four valence electrons and generally forms four covalent bonds; this is called tetravalency.

Example: C-C-C-C

Lesson 2: Organic Molecular Structures

Organic compounds can be represented using different formulas.

Molecular formula shows the number of atoms: C_2H_6 .

Structural formula shows how atoms are connected: $\text{CH}_3\text{-CH}_3$.

Condensed formula is a shorter representation: $\text{CH}_3\text{CH}_2\text{CH}_3$ represents propane.

Lesson 3: Hydrocarbons

Hydrocarbons contain only carbon and hydrogen. The main types are alkanes, alkenes, alkynes, and aromatic hydrocarbons.

Alkanes contain only single carbon-carbon bonds. General formula: $\text{C}_n\text{H}_{2n+2}$. Examples: methane (CH_4), ethane (C_2H_6), propane (C_3H_8), butane (C_4H_{10}). Alkanes are saturated hydrocarbons.

Alkenes contain at least one carbon-carbon double bond. General formula for simple open-chain alkenes: C_nH_{2n} . Examples: ethene (C_2H_4), propene (C_3H_6).

Alkynes contain at least one carbon-carbon triple bond. General formula for simple open-chain alkynes: $\text{C}_n\text{H}_{2n-2}$. Examples: ethyne (C_2H_2), propyne (C_3H_4).

Lesson 4: Homologous Series

A homologous series is a group of organic compounds with the same functional group, similar chemical properties, a common general formula, and successive members differing by CH_2 .

Example: CH_4 , C_2H_6 , C_3H_8 , C_4H_{10} .

Lesson 5: IUPAC Nomenclature

IUPAC provides standard rules for naming organic compounds.

Basic steps: find the longest carbon chain, number the chain, identify substituents, identify the main functional group, and construct the name.

Carbon prefixes: 1 Meth-, 2 Eth-, 3 Prop-, 4 But-, 5 Pent-, 6 Hex-.

Examples: $\text{CH}_4 \rightarrow$ Methane; $\text{CH}_3\text{CH}_3 \rightarrow$ Ethane; $\text{CH}_3\text{CH}_2\text{CH}_3 \rightarrow$ Propane.

Lesson 6: Functional Groups

A functional group is an atom or group of atoms that gives an organic compound its characteristic properties.

Alcohol: -OH (Ethanol)

Aldehyde: -CHO (Ethanal)

Ketone: $>\text{C}=\text{O}$ (Propanone)

Carboxylic acid: $-\text{COOH}$ (Ethanoic acid)

Ester: $-\text{COO}-$ (Ethyl ethanoate)

Amine: $-\text{NH}_2$ (Methylamine)

Lesson 7: Alcohols

Alcohols contain the hydroxyl group $-\text{OH}$. General structure: $\text{R}-\text{OH}$.

Examples: $\text{CH}_3\text{OH} \rightarrow$ Methanol; $\text{CH}_3\text{CH}_2\text{OH} \rightarrow$ Ethanol.

Alcohols are used in solvents, fuels, disinfectants, and industrial processes.

Lesson 8: Aldehydes and Ketones

Both aldehydes and ketones contain a carbonyl group, $\text{C}=\text{O}$.

Aldehydes have the carbonyl group at the end of the carbon chain. Example: CH_3CHO is ethanal.

Ketones have the carbonyl group within the carbon chain. Example: CH_3COCH_3 is propanone.

Lesson 9: Carboxylic Acids and Esters

Carboxylic acids contain $-\text{COOH}$. Example: CH_3COOH is ethanoic acid, commonly found in vinegar.

Esters contain $-\text{COO}-$. They can be produced by reacting a carboxylic acid with an alcohol:

Carboxylic acid + Alcohol $\rightleftharpoons$ Ester + Water.

Many esters have pleasant smells and are used in perfumes and flavorings.

Lesson 10: Isomerism

Isomers are compounds with the same molecular formula but different arrangements of atoms.

Structural isomers have different atom connectivity. For example, C_4H_{10} has two structural isomers: butane and 2-methylpropane.

Lesson 11: Aromatic Compounds

Aromatic compounds contain special stable ring structures. The most common example is benzene, C_6H_6 .

Benzene has six carbon atoms arranged in a ring with delocalized electrons.

Aromatic compounds are used in medicines, dyes, plastics, and chemical manufacturing.

Lesson 12: Organic Reactions

Substitution: one atom or group is replaced by another. $R-X + Y \rightarrow R-Y + X$.

Addition: atoms are added across a double or triple bond. Example: $C_2H_4 + H_2 \rightarrow C_2H_6$.

Elimination: atoms or groups are removed, often producing a double bond. $\text{Alcohol} \rightarrow \text{Alkene} + H_2O$.

Oxidation: a primary alcohol can be oxidized to an aldehyde and then to a carboxylic acid:
 $\text{Alcohol} \rightarrow \text{Aldehyde} \rightarrow \text{Carboxylic Acid}$.

Reduction can involve the addition of hydrogen: $C=C + H_2 \rightarrow C-C$.

Lesson 13: Polymers

A polymer is a large molecule formed by joining many smaller molecules called monomers.

Examples include polyethylene, PVC, nylon, and polystyrene.

Example: $n(CH_2=CH_2) \rightarrow (-CH_2-CH_2-)_n$.

Polymers are widely used in packaging, clothing, construction, and household products.

Lesson 14: Organic Chemistry in Everyday Life

Fuels: methane, propane, and other hydrocarbons are used as fuels.

Medicine: many pharmaceutical compounds contain organic functional groups.

Food: carbohydrates, fats, proteins, flavors, and preservatives contain carbon-based compounds.

Plastics: many plastics are organic polymers.

Cosmetics: organic compounds are used in perfumes, creams, and other cosmetic products.

Solved Problems

Problem 1: Classify C_3H_8 . Alkanes follow C_nH_{2n+2} . For $n=3$, $H=8$. Therefore C_3H_8 is an alkane, propane.

Problem 2: Identify the functional group in CH_3CH_2OH . It contains $-OH$, so it is an alcohol; the compound is ethanol.

Problem 3: Identify CH_3COOH . It contains $-COOH$, so it is a carboxylic acid; its name is ethanoic acid.

Problem 4: Find the molecular formula of an alkene containing five carbon atoms. Alkenes follow C_nH_{2n} , so the formula is C_5H_{10} .

Practice Exercises

1. Define organic chemistry.
2. What is catenation?
3. Why can carbon form many compounds?
4. Classify CH_4 , C_2H_4 , and C_2H_2 .
5. Write the general formula of alkanes.
6. Write the general formula of alkenes.
7. Write the general formula of alkynes.
8. What is a homologous series?
9. Name CH_3CH_3 .
10. Identify the functional group in $\text{CH}_3\text{CH}_2\text{OH}$.
11. What is the difference between an aldehyde and a ketone?
12. What is a carboxylic acid?
13. What are isomers?
14. What is benzene?
15. What is an addition reaction?
16. What is polymerization?
17. Give two examples of polymers.
18. Give three applications of organic chemistry.

Mini Project: Organic Compounds Around Us

Choose three everyday products and identify the organic compounds associated with them.

Examples: perfume, plastic bottle, fuel, food, medicine, or soap.

For each product, identify: product name; organic compound or compound class; functional group if applicable; purpose of the compound; and one everyday application.

Present the results in a table.

Final Quiz

1. What is organic chemistry?
2. Why is carbon important in organic chemistry?
3. What is catenation?
4. What are hydrocarbons?
5. What are alkanes?
6. What are alkenes?
7. What are alkynes?
8. What is the general formula of alkanes?
9. What is a saturated hydrocarbon?
10. What is an unsaturated hydrocarbon?
11. What is a homologous series?
12. What is IUPAC nomenclature?
13. What is a functional group?
14. What functional group is found in alcohols?
15. What functional group is found in carboxylic acids?
16. What is an aldehyde?

35. 17. What is a ketone?
36. 18. What are isomers?
37. 19. What is benzene?
38. 20. What is an addition reaction?
39. 21. What is a substitution reaction?
40. 22. What is oxidation?
41. 23. What is reduction?
42. 24. What is polymerization?
43. 25. Give three applications of organic chemistry.

Course Summary

The major concepts covered are carbon and catenation, covalent bonding, hydrocarbons, alkanes, alkenes, alkynes, homologous series, IUPAC nomenclature, functional groups, alcohols, aldehydes, ketones, carboxylic acids, esters, isomerism, aromatic compounds, organic reactions, polymers, and everyday applications.

Key Formulas

Alkanes: C_nH_{2n+2}

Alkenes: C_nH_{2n}

Alkynes: C_nH_{2n-2}

Alcohol: $R-OH$

Carboxylic acid: $R-COOH$

Aldehyde: $R-CHO$

Ketone: $R-CO-R'$

Ester: $R-COO-R'$